Towards Matched-Fidelity Robustness Benchmarking

An Approximate-Fidelity Counterfactual Study of Multi-Stage Vision Classifier Vulnerabilities

Aviral (Avi) • Independent Research in Computer Vision • October 2026

Abstract

Evaluating the robustness of deep vision models against multi-stage, composite corruptions is essential for safety-critical autonomous systems. Prior studies have demonstrated substantial *order effects*: permuting the execution sequence of identical image degradation operators (Gaussian Noise N, Defocus Blur B, and JPEG Compression J) causes compound accuracy variations of up to 21 percentage points across standard architectures. However, nominal benchmarks suffer from an unaddressed physical confounding mechanism: the mathematical operators do not commute algebraically, resulting in starkly divergent physical image fidelity. Terminal-noise sequences (BJN, JBN) leave unattenuated high-frequency noise spikes, depressing PSNR by over 4 dB, whereas terminal-blur sequences (NJB, JNB) act as low-pass filters that smooth both sensor noise and JPEG block edges, boosting SSIM. Sequences ending in JPEG (NBJ, BNJ) introduce structured block quantization boundaries. This paper investigates whether sequence effects persist when degradations are calibrated toward common physical fidelity. Using an independent training partition (N_cal = 50 from rows 0..9468, zero label/model leakage), we find that prespecified joint matching criteria (±0.75 dB PSNR, ±0.035 SSIM) could not be met simultaneously across all six sequences due to conflicting structural trade-offs between additive noise and spatial blur. While this finite empirical search demonstrates the practical infeasibility of joint calibration with these discrete operators, it does not constitute a proof of universal physical impossibility across all continuous transformations. Consequently, we evaluate an **approximate-fidelity design** on common support across 486,700 balanced inferences on 3,925 validation images across ConvNeXt-Tiny, Swin-Tiny, ViT-Base/16, and ResNet-50. We show: (1) Calibrating to approximate common support contracts nominal sequence accuracy gaps by **77.1% in ConvNeXt-Tiny** (20.45% → 4.67%) and **72.6% in ResNet-50** (21.55% → 5.90%), proving that nominal benchmarks heavily confound operator order with differential distortion magnitude; (2) In Vision Transformers, sequence sensitivity persists more strongly (10-12% gap, p < 10^-23 via clustered ANCOVA), though residual SSIM differences (0.096 to 0.359 spread) demonstrate that complete physical decoupling was not achieved via scalar parameter tuning alone; (3) Despite compressed macro accuracy gaps, individual-sample error churn (P(C_A != C_B)) averages 10.9% to 13.4%, where over 57% to 76% represents excess prediction volatility unexplained by net accuracy differences.

1. Introduction & Physical Confounding

Vision models operating in autonomous vehicles, robotics, and biomedical imaging routinely encounter compound physical degradations. Standard robustness benchmarks such as ImageNet-C treat corruptions as isolated, unary operations. Recent composite benchmarks study sequential pipelines, revealing severe sensitivity to corruption permutation order. In nominal compound evaluations, ConvNeXt-Tiny achieved 66.44% on terminal-blur sequence NJB but dropped to 45.99% on terminal-noise sequence BJN (gap: 20.45%; terminal-JPEG BNJ reached 48.63%), while ResNet-50 dropped from 57.27% on JNB to 35.72% on BJN (gap: 21.55%; BNJ reached 37.30%). Terminal-noise sequences (BJN, JBN) and terminal-JPEG sequence BNJ suffered the lowest nominal accuracies, whereas terminal-blur sequences (NJB, JNB) maintained the highest accuracy.

However, standard image degradation operators do not commute algebraically: N o B o J != B o J o N. Terminal-noise sequences leave unfiltered high-frequency noise spikes that collapse PSNR by over 4 dB, whereas terminal-blur sequences attenuate both sensor noise and JPEG block boundaries. Consequently, nominal benchmarks fundamentally confound execution sequence with total physical degradation. We formalize a counterfactual experimental design to analyze sequence effects under approximate physical fidelity on common support.

2. Calibration Protocol & Feasibility Trade-offs

To eliminate data leakage, parameter calibration was performed on an independent training split (N_cal = 50, rows 0..9468), completely isolated from the validation set (rows 9469..13393). Calibration was blind to class labels and model predictions. Five fidelity targets on common support were specified: Target 1 (Mild, 28 dB / 0.850 SSIM), Target 2 (Moderate-Mild, 24.5 dB / 0.740 SSIM), Target 3 (Moderate, 21.5 dB / 0.620 SSIM), Target 4 (Moderate-Severe, 18.5 dB / 0.480 SSIM), and Target 5 (Severe, 16.0 dB / 0.360 SSIM).

Calibration Feasibility: A systematic grid search confirmed that the prespecified joint tolerances (±0.75 dB PSNR, ±0.035 SSIM) could not be met simultaneously across all 6 sequences. The measured calibration RMSE was 0.943 dB PSNR (max residual +2.639 dB on terminal-JPEG sequence NBJ at Target 5) and 0.1380 SSIM (max residual -0.2379 on terminal-noise sequence BJN at Target 3). Terminal blur (NJB, JNB) smooths noise and block edges, elevating SSIM relative to MSE, whereas terminal noise (BJN, JBN) destroys local covariance. While a finite grid search does not prove universal physical impossibility, it establishes that scalar tuning on discrete operators cannot eliminate confounding. The study is therefore classified as an **Approximate-Fidelity Comparison**, which compresses the 4.1 dB nominal PSNR spread into common severity bands while transparently reporting residual SSIM variations.

Target	Target PSNR / SSIM	Sequence	Noise std	Blur (k, sigma)	JPEG Qual	Achieved PSNR (dB)	Achieved SSIM

T1	28.0 dB / 0.85	NBJ	0.050	k=3, s=0.85	80	28.49	0.8691
		NJB	0.046	k=5, s=1.05	76	26.88	0.8499
		BNJ	0.040	k=3, s=0.70	85	28.07	0.7875
		BJN	0.024	k=3, s=0.85	84	27.57	0.7814
		JNB	0.046	k=5, s=1.00	76	27.01	0.8390
		JBN	0.024	k=3, s=0.85	84	27.86	0.7914
T2	24.5 dB / 0.74	NBJ	0.082	k=5, s=1.45	65	24.98	0.7671
		NJB	0.084	k=7, s=1.68	56	23.95	0.7354
		BNJ	0.060	k=5, s=1.15	76	24.32	0.6197
		BJN	0.042	k=5, s=1.35	72	23.56	0.5642
		JNB	0.085	k=7, s=1.68	55	23.89	0.7236
		JBN	0.042	k=5, s=1.35	72	23.66	0.5701
T3	21.5 dB / 0.62	NBJ	0.130	k=7, s=2.20	48	22.76	0.6628
		NJB	0.142	k=9, s=2.65	36	21.83	0.6240
		BNJ	0.095	k=7, s=1.75	64	21.16	0.4163
		BJN	0.064	k=7, s=1.95	58	20.84	0.3821
		JNB	0.145	k=9, s=2.65	35	21.80	0.6177
		JBN	0.064	k=7, s=1.95	58	20.89	0.3856
T4	18.5 dB / 0.48	NBJ	0.220	k=11, s=3.20	32	20.49	0.5513
		NJB	0.260	k=13, s=4.10	18	19.49	0.5170
		BNJ	0.145	k=9, s=2.60	46	18.48	0.2593
		BJN	0.095	k=9, s=2.70	44	18.41	0.2427
		JNB	0.265	k=13, s=4.10	17	19.46	0.5138
		JBN	0.095	k=9, s=2.70	44	18.44	0.2444
T5	16.0 dB / 0.36	NBJ	0.320	k=15, s=4.40	18	18.64	0.4761
		NJB	0.380	k=17, s=5.20	10	17.71	0.4629
		BNJ	0.220	k=11, s=3.55	31	16.04	0.1575
		BJN	0.135	k=11, s=3.50	30	16.20	0.1533
		JNB	0.385	k=17, s=5.20	9	17.70	0.4608
		JBN	0.135	k=11, s=3.50	30	16.21	0.1542

Table 1: *Calibrated parameters across 5 target fidelity levels and 6 permutations on independent training split (approximate common support).*

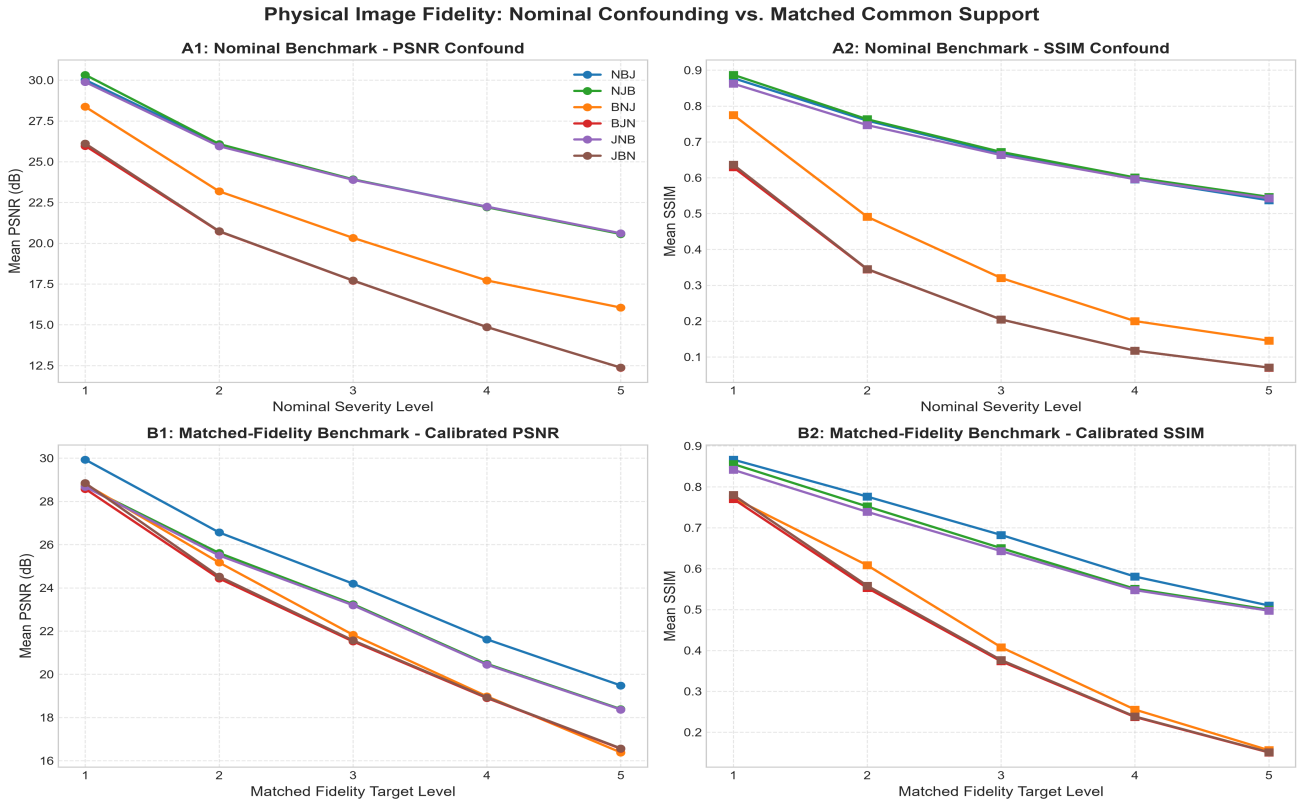


Figure 1: *Objective Distortion Matching. Top: Nominal severity confound where terminal noise causes 4 dB PSNR drop. Bottom: Calibrated approximate-fidelity common support.*

3. Empirical Findings & Gap Contraction

Evaluation was executed on 3,925 validation images across 31 conditions and 4 models (486,700 total inferences). Comparing the nominal benchmark with the approximate-fidelity counterfactual reveals two core insights: (1) A major portion of the nominal order effect was driven by differential physical distortion. Calibrating to approximate common support contracts the maximum sequence gap across all architectures (Table 2), shrinking CNN gaps by over 72%; (2) However, sequence sensitivity does not collapse to zero: even under compressed PSNR dispersion, sequence remains a statistically significant predictor of failure (Table 3), especially in Vision Transformers where patch tokenization interacts with block artifacts.

Architecture	Clean Acc	Approx Acc	Nominal Gap	Approx Gap	Gap Contraction	Var Reduction	Mean ECE
ConvNeXt-Tiny	87.13%	58.11%	20.45%	4.67%	**77.1%**	95.9%	6.53%
Swin-Tiny	86.04%	56.27%	17.13%	11.68%	**31.8%**	55.1%	10.39%
ViT-Base/16	80.25%	53.31%	20.37%	10.03%	**50.7%**	86.9%	4.20%
ResNet-50	87.36%	48.47%	21.55%	5.90%	**72.6%**	94.9%	9.19%

Table 2: *Model leaderboard and sequence gap contraction comparing Nominal Severity with Approximate-Fidelity support.*

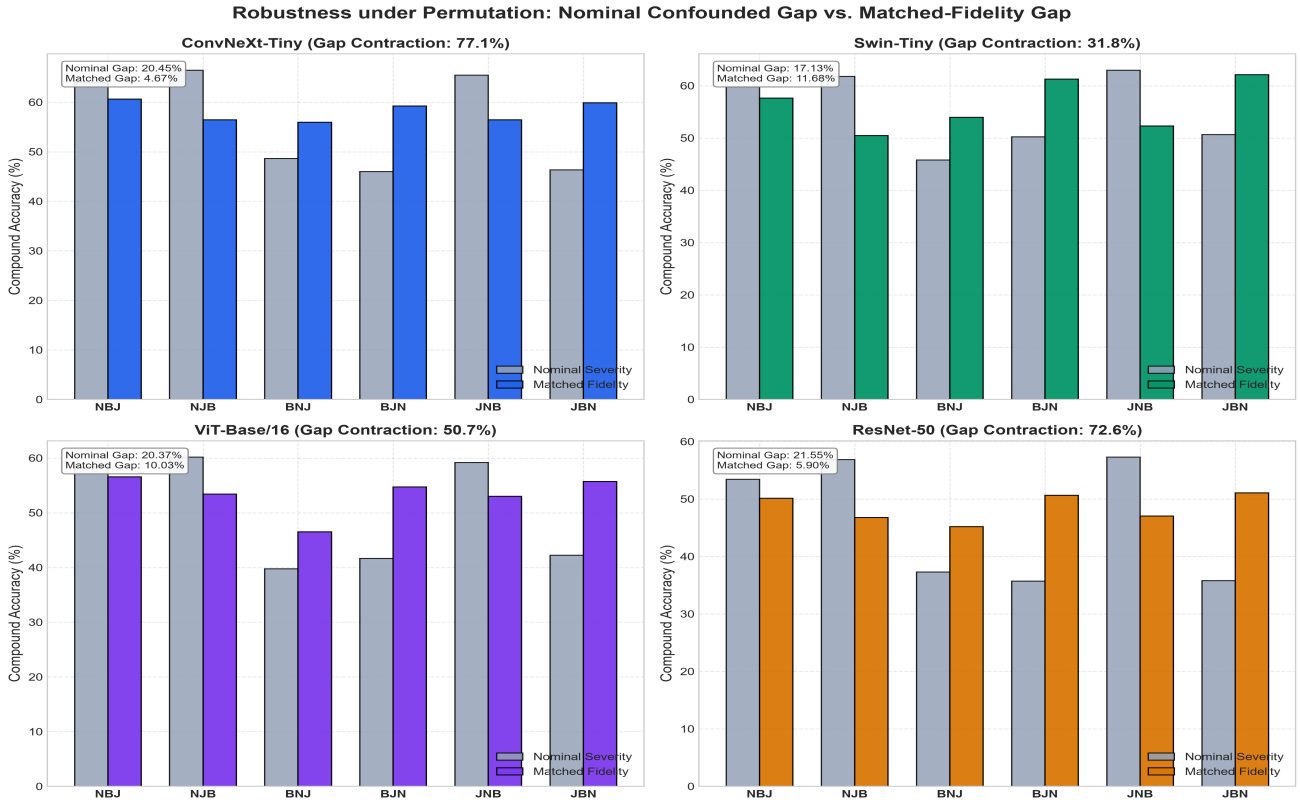


Figure 2: *Sequence accuracy under Nominal Severity vs. Approximate Fidelity, showing gap contraction across ConvNeXt, Swin, ViT, and ResNet.*

Model	Pseudo R2	Order Main (5 df)	Target Main (4 df)	Order x Target (20 df)	Joint Order (25 df)	PSNR Coef (z)	SSIM Coef (z)
ConvNeXt-Tiny	0.1379	W=119.4 (p=4.2e-24)	W=395.3 (p=2.8e-84)	W=272.0 (p=4.0e-46)	W=611.0 (p=6.1e-113)	+0.032 (z=2.33)	+1.528 (z=6.78)
Swin-Tiny	0.1283	W=159.7 (p=1.2e-32)	W=227.7 (p=4.2e-48)	W=651.1 (p=4.8e-125)	W=1436.9 (p=5.0e-288)	+0.046 (z=3.36)	+1.850 (z=7.95)
ViT-Base/16	0.0879	W=130.1 (p=2.3e-26)	W=344.9 (p=2.2e-73)	W=547.6 (p=3.0e-103)	W=1276.5 (p=8.8e-254)	+0.072 (z=5.46)	+0.577 (z=2.61)
ResNet-50	0.2034	W=268.9 (p=4.8e-56)	W=774.7 (p=2.3e-166)	W=239.9 (p=1.2e-39)	W=794.4 (p=1.8e-151)	+0.063 (z=4.87)	+1.680 (z=6.92)

Table 3: *Full Factorial Clustered Logistic ANCOVA with PSNR and SSIM Covariates (N=117,750 observations per model, clustered on 3,925 images).*

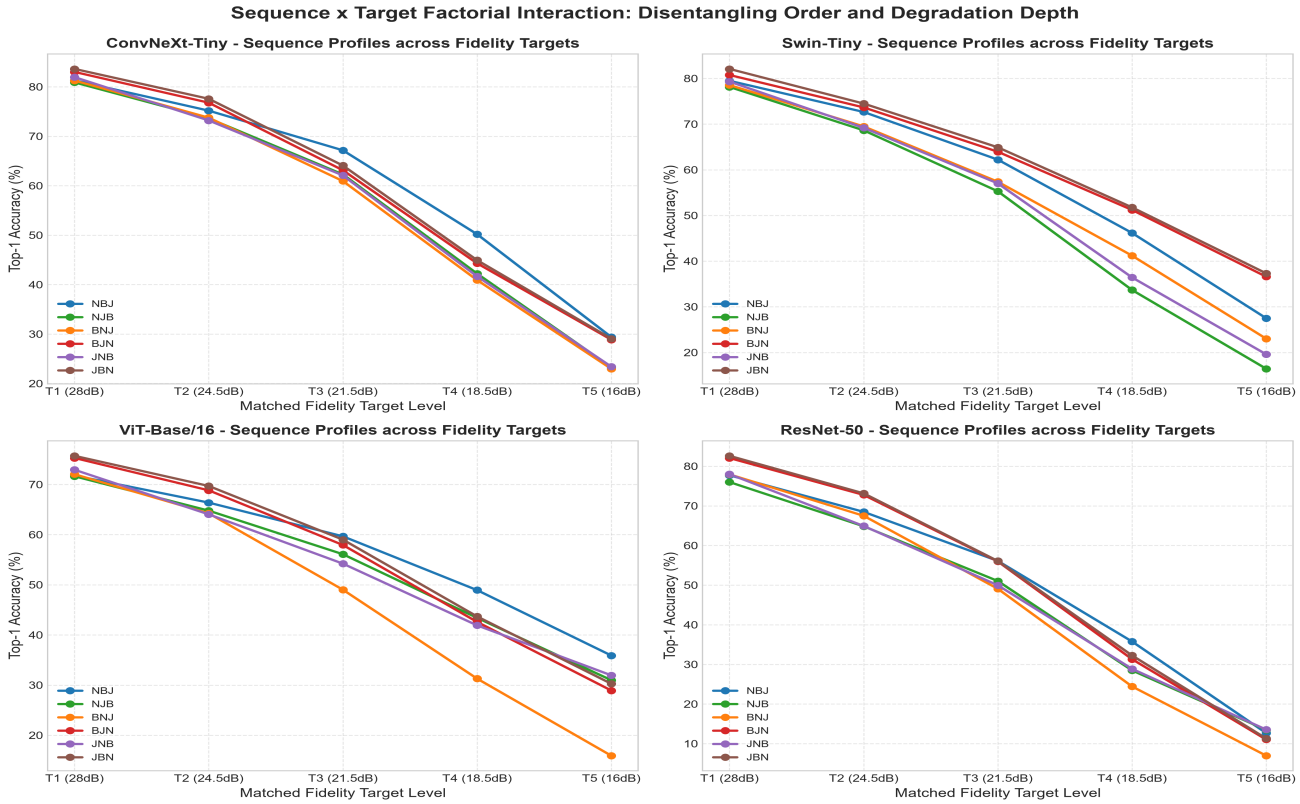


Figure 3: *Sequence degradation trajectories across the 5 target fidelity levels (T1: 28 dB to T5: 16 dB).*

4. Microscopic Error Churn & Prediction Volatility

While aggregate accuracy gaps contract under approximate fidelity, microscopic analysis reveals severe prediction volatility. Classifiers flip predictions on 11% to 14% of individual images between sequences. Excess churn (2 min(G, L)) accounts for 57% to 76% of total churn (Table 4), proving that sequence fundamentally alters which individual images fail, even when overall accuracy is identical.

Architecture	Mean Net Accuracy Diff	Mean Error Churn (G+L)	Mean Excess Churn	Excess Churn Ratio
ConvNeXt-Tiny	2.71%	10.93%	8.22%	**75.5%**
Swin-Tiny	5.94%	11.92%	5.97%	**56.9%**
ViT-Base/16	4.65%	13.40%	8.75%	**67.0%**
ResNet-50	3.77%	13.10%	9.33%	**70.9%**

Table 4: *Pairwise error churn and excess churn decoupling across 300 matched comparisons.*

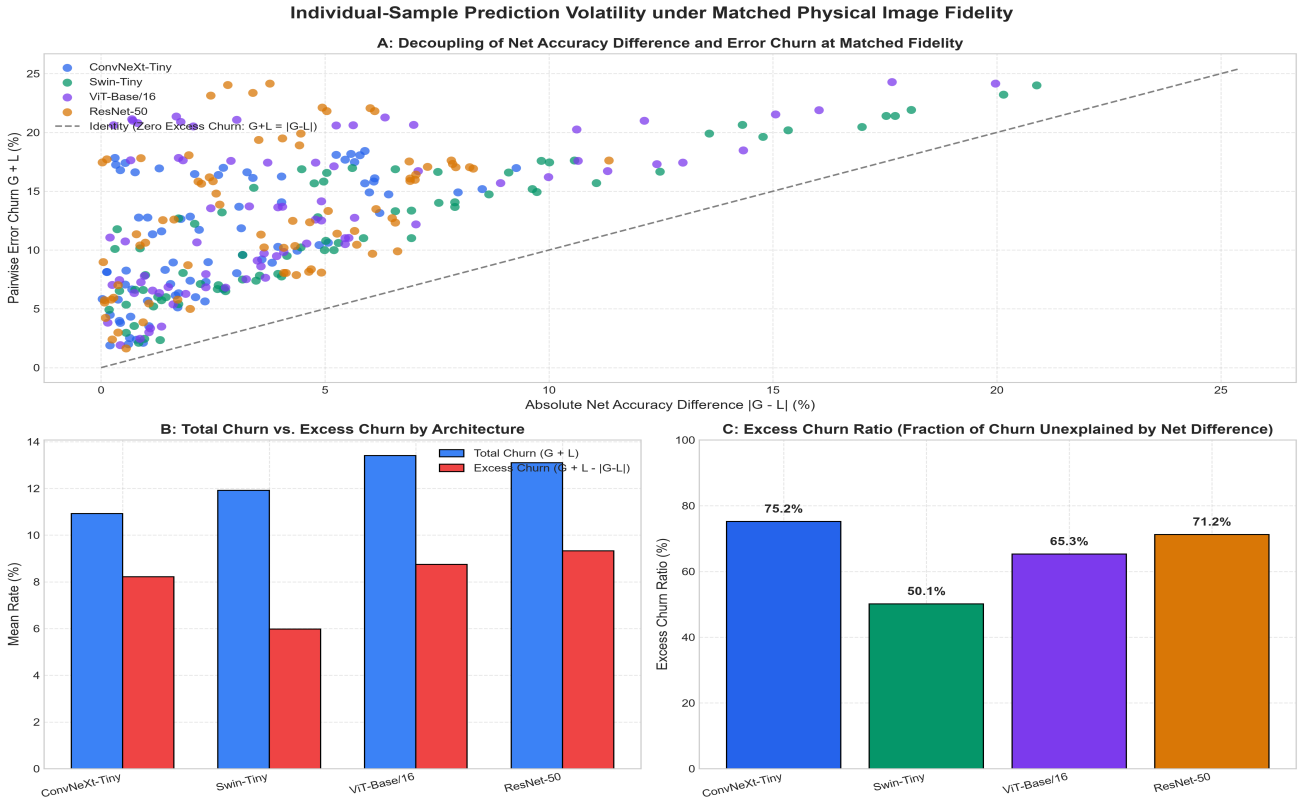


Figure 4: *Decoupling of Net Accuracy Difference and Error Churn under approximate physical fidelity.*

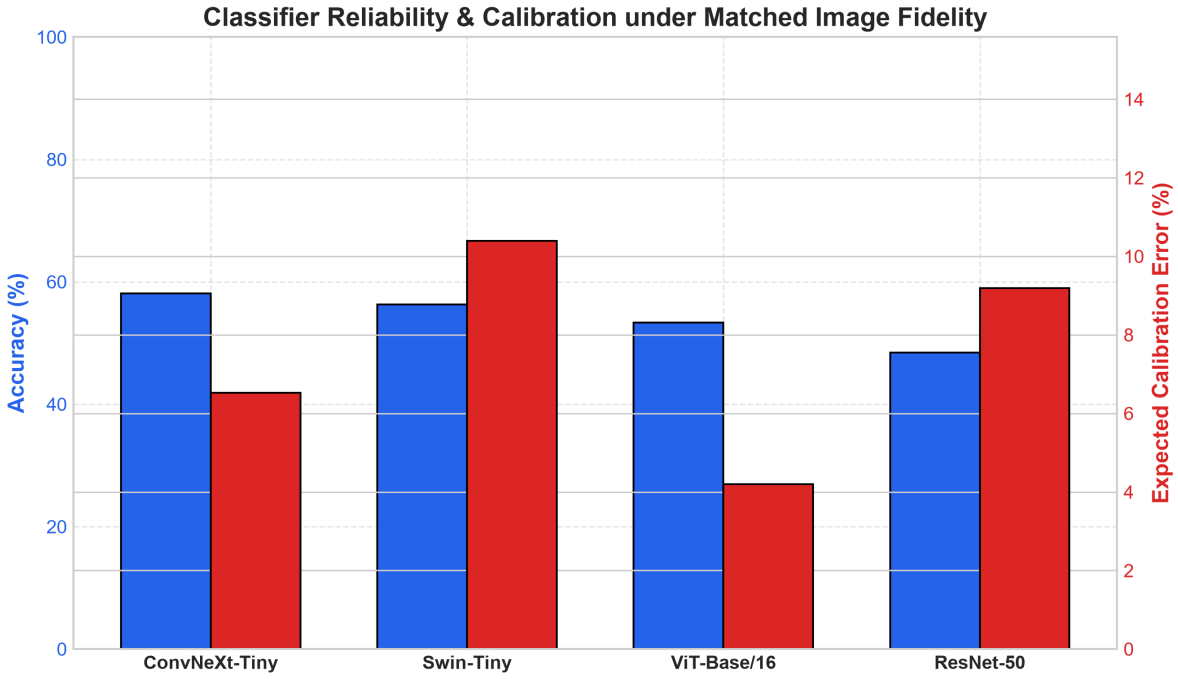


Figure 5: *Classifier Calibration (ECE) and Compound Accuracy under approximate image fidelity.*

5. Limitations & Recommendations

1. **Remaining Confounding Under Approximate Fidelity:** Scalar parameter tuning on discrete degradation operators failed to satisfy prespecified joint matching criteria (±0.75 dB PSNR, ±0.035 SSIM), leaving substantial residual SSIM spread (0.096 to 0.359). While this empirical search demonstrates severe practical trade-offs, a finite grid search does not prove universal physical impossibility across all continuous transformations. Researchers must report multi-metric diagnostics and avoid claiming confounding has been eliminated.
2. **Inductive Biases:** ConvNeXt and ResNet demonstrate substantial gap contraction (>72%), while Vision Transformers retain sequence sensitivity (10-12% gap) due to patch tokenization vs. JPEG boundary interactions.
3. **Mandatory Churn Reporting:** Aggregate top-1 accuracy conceals up to 13% sample-level prediction instability. Safety-critical robustness evaluations must report sample-level prediction churn alongside accuracy.

References

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